

Housing Prices Analysis

Executive Summary of Findings & Insights

1. Data Exploration & Cleaning Overview

The Kaggle Housing Prices dataset was analyzed to understand the primary drivers of house valuations. The dataset consisted of 545 records and 13 variables, including both numerical features (e.g., area, bedrooms) and categorical features (e.g., main road access, air conditioning).

Data quality was exceptionally high, requiring no imputation for missing values. All categorical variables were successfully transformed into numerical formats using one-hot encoding to facilitate mathematical modeling. The final prepared dataset contained 14 predictive features alongside the target variable (price), with all duplicate or invalid entries removed.

2. Predictive Modeling Performance

Two machine learning models were developed to predict house prices: a **Linear Regression** model and a **Random Forest Regressor**. Both models were trained on an 80% data split and evaluated on the remaining 20%.

- **Linear Regression:** Established a strong baseline by capturing linear relationships between features like total area and price. It proved highly interpretable and successfully mapped the base valuation trends.
- **Random Forest Regressor:** Delivered more nuanced predictive accuracy by capturing non-linear patterns and complex interactions between variables (e.g., the exponential value added by combining high square footage with air conditioning).

Overall, the models successfully explained a significant majority of the variance (R^2 score) in house prices, demonstrating that the provided features are highly predictive of real-world property valuations.

3. Key Insights & Feature Importance

Analysis of feature correlation and model feature importance revealed clear priorities for home valuation:

- **Primary Price Drivers:** **Total Area**, the **Number of Bathrooms**, and the presence of **Air Conditioning** showed the strongest positive correlation with higher house prices. These features consistently act as the strongest anchors for a home's worth.

- **Secondary Factors:** Features such as the presence of a guest room, basement, or being situated near a main road, while beneficial, had comparatively less impact on the final listing price than core structural amenities and total square footage.

4. Strategic Recommendations

"Based on the data, the most effective strategy for maximizing a property's value is investing in core structural and comfort upgrades rather than peripheral amenities."

For real estate businesses, investors, or homeowners looking to maximize Return on Investment (ROI) prior to a sale, the recommendation is clear: **prioritize adding an additional bathroom or installing central air conditioning**. These specific upgrades consistently correspond with massive jumps in final sale prices and offer the highest reliable return on investment compared to other potential home improvements.